Category-Oriented Localization Distillation for SAR Object Detection and a Unified Benchmark

Chao Wang, Rui Ruan, Zhicheng Zhao, Chenglong Li, and Jin Tang

***Abstract*— Despite much research progress in synthetic aperture radar (SAR) object detection, the performance of SAR object detection has encountered a bottleneck limited by the imaging mechanism of SAR. In this work, we investigate how to perform robust SAR object detection by distilling the category knowledge from optical images in the training stage. To this end, we propose a novel knowledge distillation (KD) method called category-oriented localization distillation (CoLD), which employs the optical object detection network as the teacher to guide the SAR object detection network. To introduce the category prior knowledge of the teacher network in the localization knowledge transferring, a category-oriented partition module (CPM) is designed in CoLD to decouple candidate bounding boxes into target and nontarget ones according to the category information in optical images. Through box decoupling, the accuracy and efficiency of SAR object detection can be significantly improved. Moreover, an intersection over union (IoU)-based weighting module (IWM) is introduced in CoLD to guide the student network focusing more on high-quality candidate boxes by adaptively changing the weight of each candidate bounding box based on the corresponding IoU score in the teacher network. In addition, a unified benchmark dataset is created for the evaluation of optical information-guided SAR object detection, which consists of 14 665 optical and SAR image pairs in the training set and 3666 SAR images in the testing set. Extensive experiments on the dataset demonstrate the effectiveness of our CoLD against state-of-the-art methods. The dataset is available at: https://github.com/mmic-lcl/Datasets-and-benchmark-code.**

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***Index Terms*— Knowledge distillation (KD), multimodal remote sensing datasets, object detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) offers persistency and endurability in challenging environments, which are widely used in military reconnaissance, exploration and guidance [1], [2], such as recognizing significant objects including bridges, harbors and oil tanks. The imaging mechanisms of SAR sensors and optical sensors are fundamentally different. SAR generates images by actively illuminating the ground objects with radio waves and receiving the backscattering energy formed by each independent scattering unit of the corresponding object. Thus, SAR images can provide more valuable information than optical images when cloud cover obstructs objects. But, the pixel value in the SAR image represents how much radar energy is reflected by the sensor, which results in the boundaries of ground objects having poor continuity and integrity. Taking the oil tank in Fig. 1 as an example, a severe substantial scattering interference exists between the object and its background, which is difficult to distinguish even for experts. Therefore, SAR object detection still continues to encounter substantial obstacles.

In recent years, SAR object detection methods [3], [4] have significantly improved performance through convolutional neural networks (CNNs) to extract more discriminative features and richer semantic information from SAR images. Despite the excellent performance of CNNs in computer vision tasks, their ability to model relationships between samples is limited, and they cannot effectively extract and represent sequential properties of spectral features. Therefore, Hong et al. propose a new minibatch graph convolutional networks (miniGCN) [5] and a new backbone network structure (SpectralFormer) [6], to improve the model's performance. The miniGCN can be trained in small batches and attempts to find better and more robust local optima. SpectralFormer reduces the loss of valuable information during hierarchical propagation to better represent the sequential properties of spectral features. In particular, ORSIm [7] and UIU-Net [8] integrate multichannel feature extraction, fast image pyramid matching and enhancement strategies, and learn the features that effectively enhance global and local contrast information. For SAR images, Du et al. [9] embed a multiscale feature attention module (MFAM) in the network to achieve a better detection performance in

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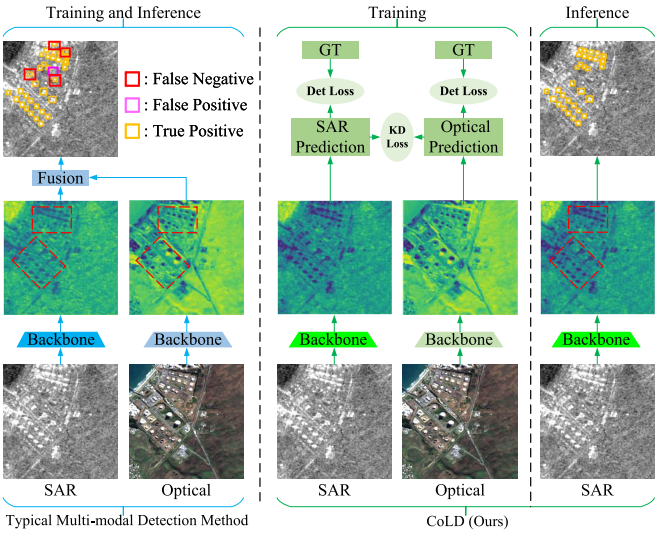


Fig. 1. Visualization results of the same regions in optical and SAR image pairs. There are correlation and complementary information between SAR and optical images. The methods based on multimodal feature fusion require pairs of images in both the training and inference stages, which greatly limits the application scene. Our proposed CoLD employs the optical object detection network as the teacher to guide the SAR object detection network. In this way, the knowledge of the optical object detection network is well transferred and only the SAR images are required in the inference stage. It is obvious that the proposed CoLD can extract more discriminative features for the objects in red dashed boxes with guidance of optical images.

SAR images under complex scenes. Cui et al. [10] densely connect the convolutional block attention module (CBAM) to each concatenated feature map to obtain rich features containing resolution and semantic information for multiscale ship detection. However, due to the limitations of the SAR imaging mechanism, the performance of SAR object detection meets a bottleneck. As shown in the first row of Fig. 1, the object detection methods have the difficulty in extracting more discriminative features from the SAR image, which leads to unsatisfactory detection results. With the rapid development of satellite technology, images of the same area captured by different types of satellites are available. Hou and Zheng [11] start to generate corresponding optical images by generative adversarial networks and then use optical images of the same region to compensate for the shortcomings of SAR object detection methods. These methods require paired data for inference, however paired data are difficult to obtain at the same time in real-world scenarios.

Cross-modal knowledge distillation (KD) is an effective solution, which requires only SAR images in the inference stage. Wang et al. [12] extract features directly from the different modalities for distillation, where features extracted from the teacher network are used to supervise the student network. Since there is a big gap between optical images and SAR images, Zhang et al. [13] extract knowledge of the optical branch by aligning the activation distributions of the two modalities. Zheng et al. [14] propose a localization distillation method to transfer location knowledge. However, the coupled Kullback–Leibler (KL) limits the efficiency of transferring candidate bounding boxes and categories. Although Zhao et al. [15] decouple KL into two parts to efficiently transfer location knowledge, it entails a high-computational cost. Cross-modal KD of optical and SAR images is a very promising task.

However, due to sensor limitations, it is difficult to acquire simultaneous optical and SAR data in realistic scenarios. In particular, there is a lack of high-quality and well-annotated public datasets. To this end, we aim to achieve robust SAR object detection by extracting category knowledge from optical images and build a unified benchmark dataset for the evaluation of optical information-guided SAR object detection. To effectively leverage the category prior knowledge in optical images, we propose a category-oriented localization distillation (CoLD) method, which improves the performance of SAR object detection by distilling the category knowledge from optical images. In specific, we design a category-oriented partition module (CPM) that separates candidate bounding boxes into target and nontarget ones using optical image category information. CPM can generate candidate bounding boxes with rich category knowledge while reducing the expensive computational costs of transferring localization knowledge. Considering that the quality of candidate bounding boxes is greatly different, we propose an intersection over union (IoU)-based weighting module (IWM) that adaptively guides the student network to focus on high-quality candidate boxes by the corresponding IoU scores in the teacher network. Our dataset, called optical image guided SAR image object detection (OGSOD-1.0), is a dataset for optical image-guided SAR image multiobject detection, where the training set consists of 14 665 optical and SAR image pairs and the testing set consists of 3666 SAR images. Extensive experiments on the dataset OGSOD-1.0 show that our method can effectively improve the performance of SAR detectors with CoLD.

In summary, the contributions of this article can be summarized as follows.

1) We propose a novel cross-modal localization distillation method called CoLD, which mines and leverages category prior knowledge to empower the student network only in the training stage. In the inference stage, we only use SAR images, this manner greatly extends the application scenario.
2) To embed category prior information efficiently, we design a CPM to obtain informative localization knowledge. The module can reduce the expensive computational costs of transferring localization knowledge while introducing category prior knowledge. Furthermore, to improve the efficiency of localization distillation, we design an IWM. This module can guide the network to focus on high-quality localization knowledge in an adaptive manner.
3) To the best of our knowledge, the OGSOD-1.0 is the first multicategory and well-annotated SAR-Optical benchmark dataset for optical image-guided SAR image in multiobject detection tasks.
4) Extensive experiments on our proposed dataset are carried out to demonstrate the effectiveness of the proposed CoLD. Compared with existing cross-modal and SAR object detection methods, the results show that our method can effectively exploit the knowledge

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in optical images to improve the performance of SAR object detection.

II. Related Work

In this section, we review the recent progress of SAR object detection methods, multimodal object detection methods, and cross-modal KD methods.

A. SAR Object Detection Methods

Traditional method constant false alarm rate (CFAR) [16] relies mainly on hand-crafted features based on object structure and scattering. CFAR method works well in simple situations. However, when the practical scenarios are more complex, it will be difficult to select a helpful model for clutter statistical distribution. Once the clutter statistical distribution [17] model is not appropriate, performance will degrade. Zhang and Liu [18] use enhanced directional smoothing (EDS) to remove speckle noise from SAR images to improve detection performance. Traditional SAR object detection methods often rely on handcrafted features, and the implementation process is complex. It is difficult to achieve good detection results in complex scenes, and its generalization ability is poor.

With the rapid advancement of deep learning theory, CNNs have been successfully applied to image classification and object detection. Deep feature descriptors obtained by CNN have gradually replaced traditional handcrafted descriptors. Numerous researchers [19], [20] advance many object detectors using the CNN framework. The raw images could be tested as inputs to generate desired prediction results as long as the network is trained with enough images and labels. When the scene becomes more complex, objects in SAR images are disturbed by noise and clutter in the background. It becomes more difficult to locate and classify. To solve this problem, Wu et al. [21] map the features of the image to the interaction space to learn the relationship between the ship and the background.

These methods mainly extract the texture and structural features, but the weak representation of this information in SAR images makes SAR object detection methods hit a bottleneck.

B. Multimodal Object Detection Methods

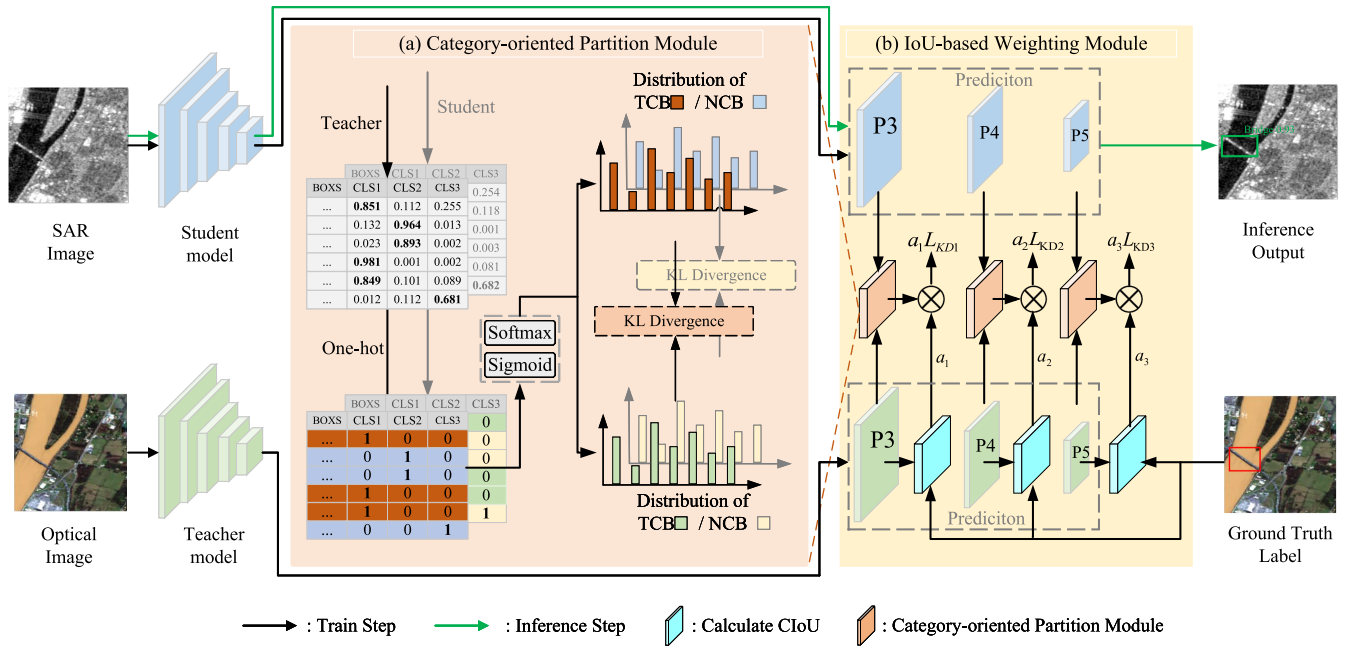
In recent years, with the rapid development of satellite technology, there exists complementarity among the information obtained by different satellites in the same area. Therefore, Hong et al. propose a shared and specific feature learning (S2FL) [22] model and a general multimodal deep learning (MDL) [23] framework. S2FL can decompose the heterogeneous data sources of remote sensing data into multishared and multispecific components, thereby enabling more effective fusion of multimodal information. Domain adaptation-based multi-source change detection network (DA-MSCDNet) [24] employs feature-level transformation to align inconsistent deep feature spaces in heterogeneous data to enhance the interaction between optical and SAR image features. Gao et al. [25] construct a plug-and-play interactive attention sum-max fusion module (IASM) to interact with features from multimodal images. However, there are significant gaps between SAR and optical images, and it is very difficult to achieve strict feature alignment due to the imaging mechanism. Hence, Shi et al. [26] propose an unsupervised domain adaptation (UDA) framework based on progressive transfer for SAR ship detection by transferring knowledge from the optical domain to the SAR domain. These methods usually require pairs of SAR-Optical data for inference, which are difficult to obtain simultaneously in real-world scenarios.

C. Cross-Modal Knowledge Distillation Methods

Early KD [27] is used for image classification and produces impressive results. Chen et al. [28] create a fast-to-execute student model to mimic a large teacher network. For object detection, knowledge transfer is more challenging than classification tasks. Early method [29] uses the features of the objects and their context as the distillation objective. Guo et al. [30] achieves better distillation performance by adjusting the weights of different locations of the feature maps. The work of [31] shows that the original Logit-based model is less efficient for object detection with low efficiency because it can only transfer some information associated with classification while ignoring other information associated with localization. Hence, Zheng et al. [14] introduce a localization distillation method, and proposes to separately transfer classification and localization knowledge based on localization regions accordingly, thereby improving distillation performance greatly. However, these methods usually focus on single-modality data and cannot adapt well to cross-modal object detection.

Recently, KD has been used for cross-modal learning. The aim of cross-modal KD is to transfer knowledge between different modalities. This approach relies more on paired data and extracts features from the same location directly from different data for distillation, where the features extracted by the teacher network are used to supervise the student network. This distillation method works well for a variety of cross-modal tasks, such as between optical and SAR images object detection [32], depth and RGB images object detection [33]. State-of-the-art distillation methods are mainly based on distilling deep features from intermediate layers, but the significance of logit distillation is greatly overlooked. Meanwhile, as the semantic knowledge and the localization are mixed on the feature maps, it is hard to tell whether it is beneficial to the performance to transfer the hybrid knowledge for each location. Our task is to use optical images to complement SAR images for object detection, thus logit mimicking-based distillation is more effective. Zheng et al. [14] prove that logit mimicking can be better than feature imitation. Zhao et al. [15] reformulate the classical KD loss into two parts, target class KD (TCKD) and nontarget class KD (NCKD), and reveal that the coupled formulation of KD limits the effectiveness and flexibility of knowledge transfer.

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Category-oriented Localization Distillation (CoLD)

Fig. 2. Overall framework of our CoLD, where TCB denotes target category bounding boxes and NCB denotes nontarget category bounding boxes. A prior knowledge of the optical images is introduced using a specially designed CPM, and candidate bounding boxes with rich knowledge are obtained. An IWM then adaptively focuses on high-quality candidate bounding boxes in the teacher network based on the corresponding IoU score. Finally, only SAR images are input for inference. (a) CPM. (b) IWM.

III. Proposed Method

The overall framework of CoLD is discussed in Section III-A. The proposed CPM and IWM are discussed in Sections III-B and III-C, respectively. The loss function of CoLD is discussed in Section III-D.

A. Overall Framework

Optical images are rich in category prior knowledge and it is critical to make full use of this knowledge to aid SAR object detection. Considering the great progress made by YOLOv5 [34] in the field of image object detection, we select the model as our baseline. Based on YOLOv5, we propose a novel CoLD to efficiently leverage different knowledge from optical images for SAR image object detection. CoLD uses optical images to train the teacher network and SAR images to train the student network, and only use SAR images to inference. As shown in Fig. 2, our CoLD mainly contains two parts: CPM and IWM.

First, to introduce the category prior knowledge of the teacher network in the localization knowledge transfer, we design the CPM. As shown in Fig. 2(a), the CPM is the core component of our CoLD. The CPM divides candidate bounding boxes into target and nontarget categories based on the optical image category. We then utilize the probability distribution representation of the bounding box to indicate the object's location. Second, to help the student network focus on high-quality bounding boxes, an IWM is designed that adaptively modifies the weight of each candidate bounding box depending on the matching IoU score in the teacher network. The structure of the IWM is shown in Fig. 2(b).

B. Category-Oriented Partition Module

Existing localization distillation methods mostly use KL-Divergence loss, which generates coupled bounding boxes. However, coupled bounding boxes can severely suppress knowledge transfer among nontarget boxes. Decoupling the bounding boxes directly can lead to expensive computational costs and make KD much less effective. And these methods do not make full use of category prior knowledge in optical images, limiting the performance of object detection.

To embed category prior information and reduce the computational costs of transferring localization knowledge, we design a CPM module. The CPM generates decoupled bounding box distributions by classifying bounding boxes into target bounding boxes and nontarget bounding boxes based on the category of the optical image. To make effective use of logit mimicking for KD, we convert the bounding box into a probability distribution [35] to better describe object localization information. As shown in Fig. 2(a), the CPM not only efficiently introduces prior knowledge of the category in the optical image but also improves the computational efficiency of transferring localization knowledge. The CPM will be discussed in detail in Sections III-B1 and III-B2.

1) Probability Distribution of the Bounding boxes: In conventional representations, there are three kinds of bounding boxes: $[x, y, w, h]$, $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$, and $[t, b, l, r]$. They employ the Dirac delta distribution solely for ground-truth (GT) locations, which cannot account for the ambiguity in bounding boxes. We represent the bounding box with a probability distribution that appropriately captures the uncertainty in its location. Assume $y \in B$ is an edge of a bounding box. The detection model may provide us with the

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expected value $\widehat{y}$

$$\widehat{y} = \int_{y_0}^{y_n} x p(x) dx \tag{1}$$

where x is the regression coordinate and $p(x)$ is the corresponding probability. Given the range of the label y with minimum y_0 and maximum $y_n (y_0 \leqslant \widehat{y} \leqslant y_n, n \in N)$, by quantizing the continuous regression range $[y_0, y_n]$ into the uniform discretized variable $[y_0, y_1, \ldots, y_n]$ with n subintervals. Given the discrete distribution property $\sum_{i=0}^{n} p(y_i) = 1$, the estimated regression value $\widehat{y}$ can be represented as

$$\widehat{y} = \sum_{i=0}^{n} p(y_i) y_i. \tag{2}$$

Thus, $p(x)$ can be easily implemented through a Softmax layer consisting of $n + 1$ units.

2) Decoupled Bounding Boxes: The probability distribution consisting of all predicted values is $p = [p(y_0), p(y_1), \ldots, p(y_t), \ldots, p(y_n)]$. We can obtain the following equation:

$$p_t = \frac{p(y_t)}{\sum_{j=0}^{n} p(y_j)} \tag{3}$$

$$p_{\setminus t} = \frac{\sum_{i=0, i\neq t}^{n} p(y_i)}{\sum_{j=0}^{n} p(y_j)} \tag{4}$$

where p_t represents target probability and $p_{\setminus t}$ represents all the other nontarget probability. Meanwhile, we declare $\hat{p}_i$ to independently model probabilities among nontarget $\hat{p} = [p_0, \ldots, p_{t-1}, p_{t+1}, \ldots, p_n]$. Each element is calculated as follows:

$$\hat{p}_i = \frac{p(y_i)}{\sum_{j=0, j\neq t}^{n} p(y_j)}. \tag{5}$$

The traditional KD method uses KL-Divergence as the loss function that can be written as

$$\begin{aligned} \mathrm{KD} &= \mathrm{KL}\left(p^T \| p^S\right) \\ &= p_t^T \log\left(\frac{p_t^T}{p_t^S}\right) + \sum_{i=0, i\neq t}^{N} p_i^T \log\left(\frac{p_i^T}{p_i^S}\right) \end{aligned} \tag{6}$$

where T is the teacher and S is the student. Comparing (3) with (4), we can find $p_i = \hat{p}_i p_{\setminus t}$, so we can rewrite (6) as follows:

$$\begin{aligned} \mathrm{KD} &= p_t^T \log\left(\frac{p_t^T}{p_t^S}\right) + \sum_{i=0, i\neq t}^{N} p_{\setminus t}^T \hat{p}_i^T \log\left(\frac{\hat{p}_i^T p_{\setminus t}^T}{\hat{p}_i^S p_{\setminus t}^S}\right) \\ &= \underbrace{p_t^T \log\left(\frac{p_t^T}{p_t^S}\right) + p_{\setminus t}^T \log\left(\frac{p_{\setminus t}^T}{p_{\setminus t}^S}\right)}_{\mathrm{KL}\left(b^T \| b^S\right)} \\ &\quad + p_{\setminus t}^T \underbrace{\sum_{i=0, i\neq t}^{N} \hat{p}_i^T \log\left(\frac{\hat{p}_i^T}{\hat{p}_i^S}\right)}_{\mathrm{KL}\left(\hat{p}_t^T \| \hat{p}_t^S\right)}, \quad \sum_{i=0, i\neq t}^{N} \hat{p}_i^T = 1. \end{aligned} \tag{7}$$

Special, $p_t + p_{\setminus t} = 1$. Equation (7) can be rewritten as

$$\mathrm{KD} = \mathrm{KL}\left(b^T \| b^S\right) + \left(1 - p_t^T\right) \mathrm{KL}\left(\hat{p}_t{}^T \| \hat{p}_t{}^S\right). \tag{8}$$

As reflected by (8), the KD loss is reformulated into a weighted sum of two terms. $\mathrm{KL}(b^T \| b^S)$ represents the similarity between the teacher's and the student's binary probabilities of the bounding box for the target. Thus, we name it target probability distribution (TPD). Meanwhile, $\mathrm{KL}(\hat{p}^T \| \hat{p}^S)$ represents the similarity between the teacher's and the student's probabilities among bounding boxes for nontarget. We name it nontarget probability distribution (NPD). Obviously, TPD can significantly affect NPD because of the presence of $(1 - p_t)$. To transfer the localization knowledge and solve the bounding boxes coupling, we introduce a hyperparameter α to replace $(1 - p_t)$. We can obtain the following calculation:

$$\begin{aligned} \mathcal{L}_{\mathrm{kd}} &= \sum_{t=0}^{N} \mathrm{KL}\left(p^T \| p^S\right) \\ &= \sum_{t=0}^{N} \left[\mathrm{KL}\left(b^T \| b^S\right) + \alpha \mathrm{KL}\left(\hat{p}_t^T \| \hat{p}_t^S\right)\right] \end{aligned} \tag{9}$$

where $\mathcal{L}_{\mathrm{kd}}$ denotes the KD loss after decoupling.

Compared to KD, this strategy clearly requires N times more computation. The greater N (in experiments, N is often set to 1000), the more calculations. The above method (166.76 h) takes about four times longer to train than KD (42.12 h) and LD (41.50 h), which significantly increases training time and weakening the benefit of logit distillation. To solve this problem, we propose a method to partition the bounding boxes according to the category prior knowledge of the teacher network. As shown in Fig. 2(a), we employ the one-hot method to convert classification predictions into hard labels in the classification branch of the model. The regression branch is then guided by this hard label to split the bounding boxes. Finally, we use Softmax$(\cdot, \tau)$ to convert the bounding boxes into a probability distribution. The probability distribution may provide more valuable information when $\tau > 1$. Thus, (9) can be rewritten as follows:

$$\begin{aligned} \mathcal{L}_{\mathrm{kd}} &= \mathrm{KL}\left(m^T \| m^S\right) \\ &= \sum_{t=0}^{c} \left[\mathrm{KL}\left(m_t^T \| m_t^S\right) + \alpha \mathrm{KL}\left(\hat{m}_t{}^T \| \hat{m}_t{}^S\right)\right] \end{aligned} \tag{10}$$

where c indicates the number of categories, m_t denotes the set of all probability distributions belonging to the tth category, and $\hat{m}_t$ denotes the set of all probability distributions with category t removed.

C. IoU-Based Weighting Module

Since the resolution and imaging mechanism of the two modalities is different, the candidate bounding boxes of the teacher network are also not accurate. Existing methods tend to learn all candidate bounding boxes equally, which results in the inference of student networks being affected by low-quality candidate boxes.

To help the student network focus on high-quality candidate bounding boxes, an IWM is designed that adaptively modifies the weight of each candidate bounding box depending on the matching IoU score in the teacher network. The IWM gives

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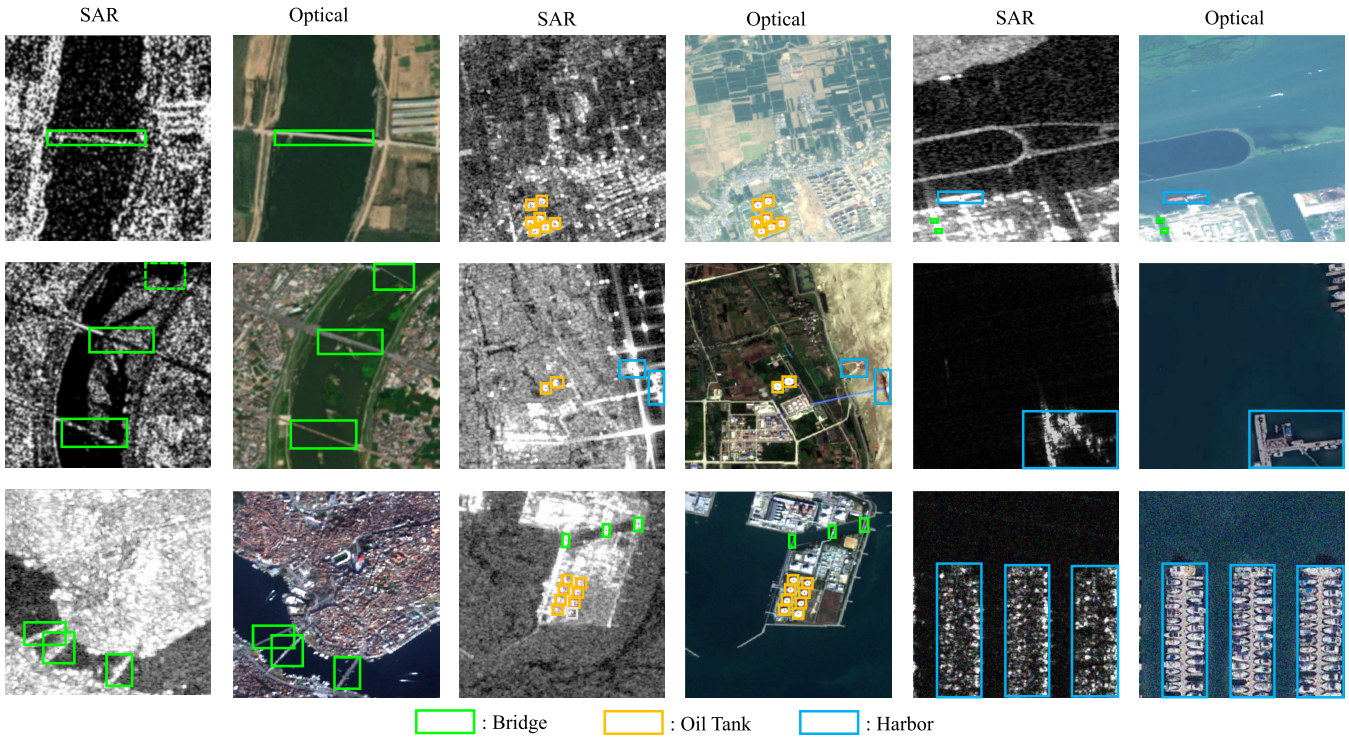


Fig. 3. Visualization of optical and SAR image pairs in OGSOD-1.0, where bridges, harbors, and oil tanks are indicated by green, blue, and yellow boxes, respectively. Despite using static objects for annotation, images from various time periods still include missing objects (e.g., the green dashed box in the SAR image), revealing our dataset is more challenging.

higher weights to high-quality candidate bounding boxes. The IWM improves the efficiency of localization distillation by effectively focusing on high-quality candidate bounding boxes, thus robustly improving the performance of student networks. As shown in Fig. 2(b), we first calculate the IoU [36] scores for the different network layer candidate bounding boxes in the teacher network. The IoU score a_i of layer i (three network layers of YOLOv5 inference, $1 \leqslant i \leqslant 3$) is calculated as follows:

$$a_i = \mathrm{IoU}\left(B_i^T, B^G\right) \tag{11}$$

where B_i^T indicates the bounding boxes of layer i of the teacher network output, B^G indicates GT boxes. Then, the IoU score a_i is used as the weight of the current candidate bounding box to focus on high-quality bounding boxes. The final distillation loss is as follows:

$$\mathcal{L}_{\mathrm{kd}} = \sum_{i=1}^{3} a_i L_{\mathrm{kd}}\left(B_i^T \| B_i^S\right) \tag{12}$$

where B_i^T and B_i^S denote the candidate bounding boxes for the teacher and the student network outputs, respectively.

D. Loss Function

The inputs of CoLD are optical images and SAR images. The optical images are first fed into the teacher network and the different category prior knowledge is extracted and fed into the CPM. Then, the SAR images are fed into the student network. After processing by the CPM, the student network generates candidate bounding boxes with rich knowledge. Finally, the IWM applies an adaptive focusing on high-quality knowledge. The entire network is trained using online KD (OKD) and the teacher network guides the student network as they train. The teacher loss is calculated as follows:

$$\mathcal{L}_T(\Theta_1) = \mathcal{L}_{\mathrm{cls}}\left(C^T \| C^G\right) + \mathcal{L}_{\mathrm{reg}}\left(B^T \| B^G\right) \tag{13}$$

where Θ_1 denotes the parameter set of the teacher network, C^T denotes categories of the teacher network output, and C^G denotes the corresponding GT. The student loss is calculated as follows:

$$\mathcal{L}_S(\Theta_2) = \mathcal{L}_{\mathrm{cls}}\left(C^T \| C^G\right) + \mathcal{L}_{\mathrm{reg}}\left(B^T \| B^G\right) + \mathcal{L}_{\mathrm{kd}}\left(B^T \| B^S\right) \tag{14}$$

where Θ_2 denotes the parameter set of the student network. Here we use complete intersection over union (CIoU) [36] since it gives higher priority to the locations closer to the object's center. Rather than simply discarding low-quality bounding boxes, the IWM dynamically adjusts the contribution of bounding boxes through weighting. Because the missing teacher knowledge is not representative of the true distribution of the data, which is detrimental to the performance of the detector.

IV. Dataset

In this section, we will introduce our OGSOD-1.0 dataset including existing datasets, dataset construction, and dataset challenges.

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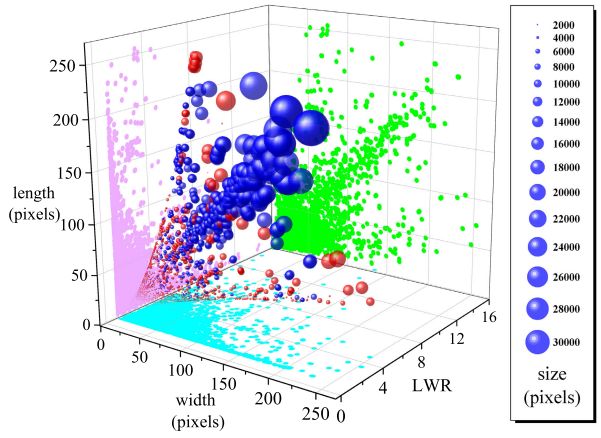


Fig. 4. Statistical distribution of object sizes in OGSOD-1.0. Bridges, harbors, and oil tanks are shown in red, blue, and yellow, respectively. LWR indicates length-to-width ratio of the object's bounding box.

A. Existing Datasets

Zhang et al. [13] construct an oil tank detection dataset for the study of optical image-guided SAR image object detection. While this dataset contains 3401 pairs of full-polarized SAR images and corresponding optical images. Only 820 pairs of images containing oil tanks are selected for labeling. Furthermore, only 18 000+ tank instances are labeled. Therefore, we propose a new dataset for optical image-guided SAR image object detection tasks.

B. Dataset Construction

The SAR images in OGSOD-1.0 are collected from the Chinese GaoFen-3 satellite in C-band, Vertical-Vertical (VV) and Vertical-Horizontal (VH) polarization modes. These SAR images are provided by the 38th Research Institute, China Electronics Technology Group Corporation, and the resolution of SAR images is 3 m. The optical images are provided by Google Earth, and the resolution is 10 m.

For the training set, aligning moving objects in the images is an obstacle due to temporal phase inconsistencies, and this can significantly affect the performance of object detection. Therefore, we obtain the optical images of the corresponding locations on Google Earth with a time difference of less than one month according to the latitude, longitude and imaging time provided by the SAR images. We map the optical and SAR images to the same spatial resolution using the ENVI software and then align them. Then, we crop the image pairs into 256 × 256 pixel slices. To mitigate the effects of moving objects, we annotate static objects, such as bridges, harbors and oil tanks. In addition, to increase the diversity of the training set, we obtain Michael Schmitt's consent to extend our dataset by selecting an additional 3000 sample pairs from the SEN1-2 [37] dataset. Therefore, the training set contains 14 665 optical and SAR image pairs. Examples of image pairs are shown in Fig. 3, these objects are well-identified and informative in the optical image, whereas they are difficult for a nonexpert to identify in the SAR image.

To simulate the scenario of lacking simultaneous phase data during inference, we construct a test set with 3666 SAR images, with the size of 256 × 256 pixel slices. Therefore, the OGSOD-1.0 consists of a training set of 14 665 optical and SAR image pairs and a test set of 3666 SAR-only images, containing over 48 000 instance annotations in total. The distributions of instance scales in the OGSOD-1.0 dataset are shown in Fig. 4, which shows that although there are multiscale objects, most of them are concentrated at smaller scales. This indicates that our dataset OGSOD-1.0 has a significant challenge.

C. Dataset Challenges

Although the OGSOD-1.0 dataset is manually aligned and screened, there are still many challenges. In summary, the difficulties in OGSOD-1.0 are mainly divided into three aspects.

1) Due to the different imaging times and imaging resolutions of satellites, some spatial transformations must be performed during alignment. This can lose some of the original information of the data and interfere with the extraction of object features. Besides, the presence of moving objects can exacerbate this situation.
2) SAR images have more noise than optical images due to the satellite's different incident angles, imaging mechanisms and environmental factors. While using optical images to guide SAR image detection often introduces clutter noise information to interfere with object detection.
3) The large variation in scale of objects in the dataset which contains many small objects. Especially the similarity of the background to the objects increases the challenge of object detection.

In addition, we select images with strong background noise interference to artificially increase the challenges of our dataset.

V. Experiments and Discussion

In this section, the proposed CoLD and other state-of-the-art methods are comprehensively evaluated on the OGSOD-1.0 dataset. First, the implementation details of the CoLD are described in Section V-A, and the evaluation metrics are described in Section V-B. Then, the effectiveness of our CoLD is investigated in Section V-C, ablation experiments and visualizations are reported in Section V-D. Finally, we compare advanced oriented object detection methods in Section V-E.

A. Implementation Details

Our method is trained on a platform with an NVIDIA RTX 3090 GPU, and the implementation is based on the public platform PyTorch [47]. In our proposed CoLD, we generally set the batch size to 64, the image size to 256 × 256 and the epoch to 400. YOLOv5 is modified to be a two-branch cross-modal teacher–student network while ensuring the same data enhancement for both optical and SAR images. For the teacher network input branch, we consistently use optical images. In the training stage, we use optical and SAR images to train the optical branch and the SAR branch separately, while in the inference stage only the SAR images are input for inference. To be consistent with the baseline and other SAR object detection methods, we set input image size as 256 × 256.

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TABLE I

COMPARISON WITH THE STATE-OF-THE-ART OBJECT DETECTION METHODS ON OUR PROPOSED OGSOD-1.0 DATASET WITH SAR IMAGES. C AND S DENOTE CROSS-MODAL METHODS AND SINGLE-MODAL METHODS, RESPECTIVELY. HBB DENOTES HORIZONTAL BOUNDING BOX

HBB Method	Venue	Backbone	Type	C/S	Parameters	GPUHour	OGSOD-1.0 (SAR HBB)				
							Oil Tank	Bridge	Harbor	AP_{50}	AP
RetinaNet [38]	ICCV'2017	ResNet101	Anchor-based	S	55.14M	34.50	17.3	73.3	95.3	62.0	36.7
YOLOv3 [39]	arXiv'2018	DarkNet53	Anchor-based	S	61.53M	40.87	32.4	76.0	97.0	68.5	39.5
RepPoints [40]	ICCV'2019	ResNet101	Anchor-free	S	36.60M	37.00	26.2	78.0	96.3	66.8	38.6
ATSS [41]	CVPR'2020	ResNet101	Anchor-based	S	50.88M	27.20	30.4	70.8	95.4	65.5	37.9
YOLOv5 [34]	-'2020	CSPDarkNet-X	Anchor-based	S	86.23M	41.11	57.7	87.2	97.9	80.9	46.3
Generalized Focal [42]	NeurIPS'2020	ResNet101	Anchor-based	S	51.03M	36.90	33.4	72.8	96.5	67.6	41.8
Sparse R-CNN [43]	CVPR'2021	ResNet101	Anchor-free	S	124.94M	90.65	28.7	73.8	94.2	65.6	38.7
Object_Box [44]	ECCV'2022	CSPDarkNet-X	Anchor-free	S	86.19M	39.68	51.0	82.4	96.5	76.6	40.1
YOLOv7 [45]	arXiv'2022	CSPDarkNet-E6	Anchor-based	S	97.20M	37.15	59.7	79.8	98.1	79.2	45.1
KD [27]	TIP'2020	CSPDarkNet-X	Anchor-based	C	86.24M	42.12	60.3	88.4	98.8	82.6	48.4
GI Imitation [46]	CVPR'2021	CSPDarkNet-X	Anchor-based	C	86.33M	67.50	<u>69.2</u>	<u>92.9</u>	<u>99.1</u>	<u>87.1</u>	<u>55.9</u>
DKD [15]	CVPR'2022	CSPDarkNet-X	Anchor-based	C	86.13M	43.17	62.5	89.1	98.7	83.4	49.8
LD [14]	CVPR'2022	CSPDarkNet-X	Anchor-based	C	86.25M	41.50	65.7	90.1	98.3	84.5	51.9
CoLD (Ours)	-	CSPDarkNet-X	Anchor-based	C	86.23M	42.59	**69.8**	**93.5**	**99.5**	**87.6**	**56.7**

[1] **Best** and <u>*Second best*</u>, Higher AP_{50} and AP are better.

We use a stochastic gradient descent (SGD) optimizer with an initial learning rate of 10^{-2}, a momentum of 0.937, and a weight decay of 5×10^{-4}. Then the entire network is trained in an end-to-end manner. For data enhancement, we have used Mosaic [48] and Mixup [49]. To ensure fairness, we set uniform settings for training and inference of other hyperparameters, such as temperature $\tau = 20$ in $\text{SoftMax}(\cdot, \tau)$, $\alpha = 2$.

B. Evaluation Metrics

All models are evaluated with two object detection metrics: mean average precision (AP) and AP_{50}

$$\text{AP} = \frac{1}{n}\sum_{i=0}^{n}\text{AP}_i = \frac{1}{n}\sum_{i=0}^{n}\int_0^1 P_i(r)dr \tag{15}$$

$$\begin{aligned}\text{AP}_i &= \int_0^1 P_i(r)dr \\ &= \int_0^1 \text{Precision}\; d(\text{Recall}) \\ &= \int_0^1 \frac{\text{TP}}{\text{TP}+\text{FP}} d\left(\frac{\text{TP}}{\text{TP}+\text{FN}}\right).\end{aligned} \tag{16}$$

When IoU value between a predicted box and the GT meets the predefined threshold, this box is represented as true positive (TP). Otherwise, this box is considered as a false positive (FP). False negative (FN) means there is a true target, but the detector does not find it. Equation (16) indicates that AP is the integral of the precision-recall curve (PRC) for each category. AP_{50} computes the mean of all the AP values for all categories at IoU = 0.50 in (15). Similarly, AP is the primary challenge metric, which can be formulated as the mean at IoU = 0.50 : 0.05 : 0.95.

C. Quantitative and Qualitative Evaluation

To verify the effectiveness of our CoLD, we conduct optical image-guided SAR image object detection experiments on our proposed OGSOD-1.0 dataset. We report the results with nine state-of-the-art SAR object detection methods and four cross-modal KD methods, including anchor-based models YOLOv3 [39], YOLOv5 [34] and YOLOv7 [45], anchor-free models Sparse R-CNN [43], RetinaNet [38], adaptive training sample selection (ATSS) [41], representative points (RepPoints) [40], Generalized Focal [42] and Object_Box [44], and cross-modal KD methods KD [27], decoupled knowledge distillation (DKD) [15], LD [14], and general instance (GI) Imitation [46].

As shown in Table I, our method CoLD achieves an improvement of 10.4% and 0.8% with optical image guidance compared to SAR object detection methods and cross-modal KD methods. Different SAR models vary considerably in terms of performance, mainly because some methods are not applicable to SAR images. YOLOv5 performs well on evaluation metrics among the SAR models due to better generalization capabilities of YOLOv5, especially for small objects. Among the cross-modal KD methods, KD and DKD transfer only class knowledge from optical images, resulting in poor accuracy. Localization knowledge is particularly important in object detection. Compared to KD and DKD, LD and GI Imitation transfer object localization knowledge in optical images with better results. LD transfers localization knowledge through the probability distribution of the bounding box and GI Imitation transfers localization knowledge by minimizing the feature similarity of the region of interest. By introducing category prior knowledge from optical images and leveraging the localization knowledge of the teacher network, our CoLD achieves the best performance.

Our model not only achieves good performance on the evaluation metrics AP but also gains significant improvements in perceptual quality. As shown in Fig. 5, our CoLD can obtain more guidance information from different knowledge in the optical images, and reduce missed and our model achieves good performance detection rates. Especially for small objects, our method achieves a huge improvement in

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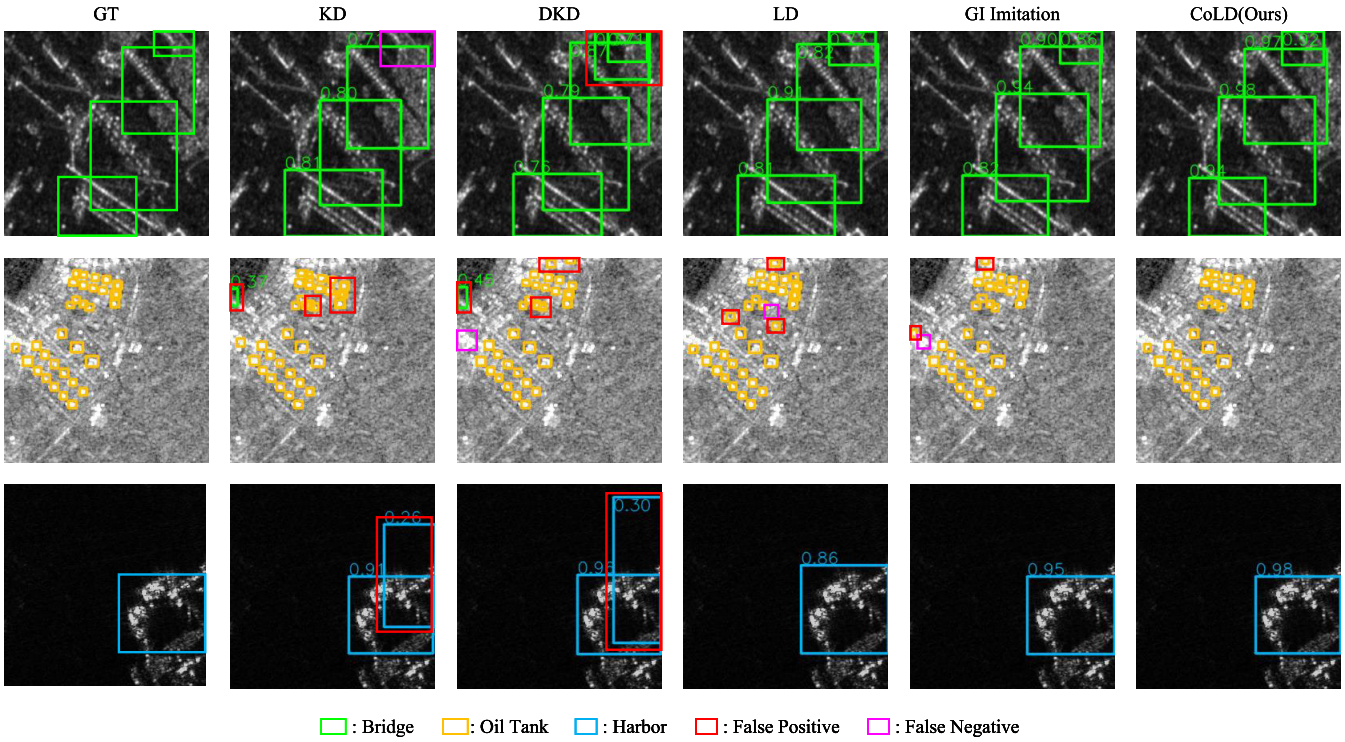


Fig. 5. Cross-modal KD strategies are visually compared. KD and DKD do not employ localization information, which leads to occasionally missed detections, while LD improves the results somewhat but falls short of GI Imitation. Our method is more accurate and has fewer missed detections according to the inference result. The red and purple boxes indicate false positive and false negative, respectively.

TABLE II

TRAINING RESULTS FOR MIXED DATA, USING ONLY SAR IMAGES IN THE INFERENCE STAGE. MIX OPTICAL-SAR REPRESENTS A MIXTURE OF OPTICAL IMAGES AND SAR IMAGES

Training Data (HBB)	Oil Tank	Bridge	Harbor	AP_{50}	AP
SAR	**57.7**	87.2	97.9	80.9	46.3
Mix Optical-SAR	56.6	**87.7**	**99.1**	**81.1**	**48.3**

SAR images. Even with a large amount of clutter noise in the SAR images, our model performs well when guided by the optical images. By contrast, there are missed detections for KD, DKD, and LD, and error detections for KD, DKD, LD, and GI Imitation. The major reason for the low accuracy of KD and DKD is the localization knowledge in the optical image that is ignored and LD fails to consider the coupling of bounding boxes. These methods make insufficient use of the optical image guidance information, which makes them detect unsatisfactory objects. The visualization results illustrate that CoLD can obtain more guidance information from the optical images and therefore can better detect objects.

To further confirm the efficacy of our strategy, as shown in Fig. 6, we visualize the learning attention map using GradCAM. YOLOv5 does not focus effectively on smaller bridges and oil tanks, as seen by the visualization results. While DKD focuses on the targets more thoroughly, it also focuses on a great quantity of background noise. Compared with other methods, our CoLD improves object focus and reduces background noise, outperforming the feature-based method of GI Imitation.

Compared with cross-modal KD methods, the number of parameters of our proposed method is comparable. As shown in Fig. 7(b), for the training time, KD, DKD, and LD use less time but achieve less accuracy, while GI Imitation achieves good accuracy with a high training cost. CoLD increases a little of training time but achieves the highest accuracy. Given the huge improvement in accuracy, the increase in the number of parameters and training time is acceptable. Our CoLD makes a great trade-off between consumption and performance.

To verify the generality of our method, we have implemented our method on Object_Box [44]. As shown in Fig. 7(c), our method is effective in improving the performance of the student model. In particular, in a multimodal method, we must assess if the CoLD's efficacy is brought about by training with mixed data. As shown in Table II, training with mixed data does not increase the accuracy of detection. It validates that the improvement is not derived from training with mixed data but from our method's efficacy.

D. Ablation Experiments

1) Effectiveness of TCLD and NCLD: To validate the capability of our designed CPM to mine guidance information, we divide all bounding boxes into two parts, each called target category localization distillation (TCLD) and nontarget category localization distillation (NCLD). As shown in Table III, TCLD gets an improvement of 3.9% (from 46.3% to 50.2%) in AP, and NCLD gets an improvement of 10.0% (from 46.3% to 56.3%) in AP. Intuitively, TCLD concentrates on the knowledge related to the target category since the corresponding loss function considers only binary probabilities. Conversely, NCLD focuses on the knowledge among nontarget categories. We notice that singly applying

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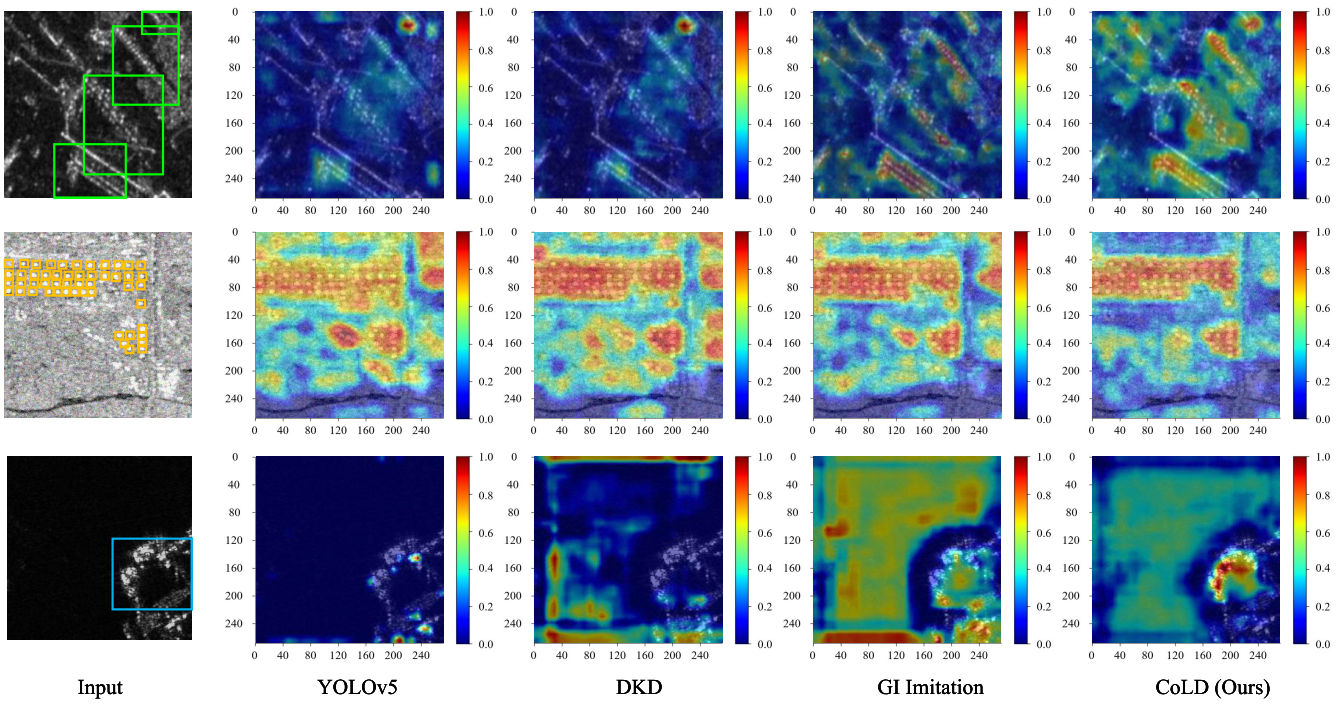


Fig. 6. Visualization of the learning attention map using GradCAM. The results show that our method allows for a better focus on the position of the object and has very little background noise. The SAR object detection method cannot effectively focus on the location of the oil tanks without the guidance of optical images. Cross-modal methods do not take full advantage of the information in optical images, resulting in the introduction of more background noise.

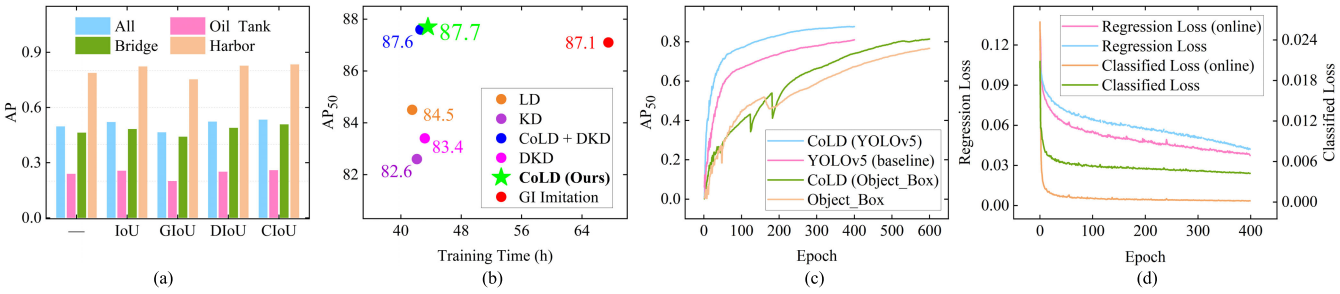


Fig. 7. (a) Performance comparison of using different IoUs (e.g., IoU, CIoU, DIoU, or GIoU) in IWM. (b) Training time (h) versus accuracy (AP_{50}) on OGSOD-1.0. The figure shows the accuracy and training time for each method. (c) Performance of CoLD in different methods. The results show that our method is strongly generalizable. (d) Performance comparison between online and offline distillation. The results show that online distillation allows the network to converge faster and gives lower loss values.

TCLD could be less helpful for the student. However, the distillation performance of NCLD is comparable and even better than the LD. According to (8), TCLD transfers knowledge via the binary classification task, which could be related to the difficulty of the sample. For instance, a training sample with $p_t = 0.95$ could be easier for the student to learn compared with another one with $p_t = 0.63$. Therefore, we use α to replace $1 - p_t$, giving a 10.4% (from 46.3% to 56.7%) improvement to the model.

2) Effectiveness of IWM, OKD, and CPM: To research the performance differences of different IoU in IWM modules, we use different IoU methods [e.g., CIoU, distance intersection over union (DIoU), or generalized intersection over union (GIoU)] to weight the distillation loss. The results are shown in Fig. 7(a). Compared to the baseline, our proposed IWM achieves the highest improvement with the introduction of CIoU. This result proves that there is low-quality knowledge in the teacher network. Therefore, we need to use a strategy to focus on high-quality knowledge.

TABLE III

ABLATION EXPERIMENTS WITH TCLD AND NCLD. TCLD INDICATES TARGET CATEGORY LOCATION DISTILLATION, AND NCLD INDICATES NONTARGET CATEGORY LOCATION DISTILLATION. THE ABLATION RESULTS (HBB) SUGGEST THAT TCLD KNOWLEDGE COULD NOT BE AS IMPORTANT AS KNOWLEDGE AMONG NCLD

TCLD	NCLD	Oil Tank	Bridge	Harbor	AP_{50}	AP
-	-	57.7	87.2	97.9	80.9	46.3
✓		71.5	82.4	91.3	81.7	50.2
	✓	68.8	93.3	94.4	87.2	56.3
✓	✓	**69.8**	**93.5**	**99.5**	**87.6**	**56.7**

To further improve the efficiency of the teacher network in transferring knowledge, we analyze OKD in depth. The experimental results are shown in Fig. 7(d), student models can converge faster in classification and regression. Offline KD entails using a pretrained teacher network to transfer learned knowledge to the student network. However, this method

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TABLE IV

ABLATION EXPERIMENTS OF OKD, CPM, AND IWM ON THE OGSOD-1.0 (HBB). OKD INDICATES ONLINE KD. CPM AND IWM DENOTE CATEGORY-ORIENTED PARTITION MODULE AND IOU-BASED WEIGHTING MODULE, RESPECTIVELY

CPM	IWM	OKD	Oil Tank	Bridge	Harbor	AP_{50}	AP
-	-	-	57.7	87.2	97.9	80.9	46.3
✓			62.8	89.4	98.8	83.7	49.7
		✓	65.7	90.1	98.3	84.5	51.9
✓	✓		66.5	91.8	99.1	85.7	53.4
✓	✓	✓	**69.8**	**93.5**	**99.5**	**87.7**	**56.7**

TABLE V

EXPERIMENTAL RESULTS OF DIFFERENT COMBINATIONS OF KD METHODS (HBB). ALTHOUGH OUR METHOD CoLD ONLY DISTILLS KNOWLEDGE THROUGH THE REGRESSION BRANCH, IT CAN EFFECTIVELY INTRODUCE CATEGORY PRIOR KNOWLEDGE

KD	DKD	LD	CoLD	Oil Tank	Bridge	Harbor	AP_{50}	AP
-	-	-	-	57.7	87.2	97.9	80.9	46.3
✓				60.3	88.4	98.8	82.6	48.4
	✓			62.5	89.1	98.7	83.4	49.8
		✓		65.7	90.1	98.3	84.5	51.9
			✓	69.8	**93.5**	**99.5**	87.6	**56.7**
✓		✓		68.5	92.6	99.0	86.7	55.8
	✓	✓		69.0	93.0	99.4	87.1	56.2
✓			✓	69.5	93.3	99.4	87.4	56.0
	✓		✓	**70.2**	93.4	99.4	**87.7**	56.6

TABLE VI

EXPERIMENTAL RESULTS OF OPTICAL IMAGES AND SAR IMAGES OF CoLD UNDER DIFFERENT CONDITIONS ON OUR PROPOSED OSPRC DATASET. SAR-VV AND SAR-VH ARE TWO DIFFERENT POLARIZATION MODES OF SAR IMAGING. OPT-10M AND OPT-20M REPRESENT OPTICAL IMAGES WITH DIFFERENT SPATIAL RESOLUTIONS, WHILE OPT-CLOUD REPRESENTS AN OPTICAL IMAGE WITH CLOUD OCCLUSION

OSPRC (HBB)					IWM	AP_{50}
SAR-VV	SAR-VH	OPT-10m	OPT-20m	OPT-Cloud		
Single Image						
✓						59.4
	✓					60.2
		✓				70.3
			✓			51.9
				✓		41.0
OPT Guided SAR						
✓		✓			✓	64.1 (+ **4.7**)
✓			✓		✓	62.0 (+ **2.6**)
✓				✓	✓	58.5 (− **0.9**)
✓				✓		51.5 (− **7.9**)
	✓	✓			✓	65.3 (+ **5.1**)
	✓		✓		✓	60.1 (− **0.1**)
	✓			✓	✓	59.3 (− **0.9**)
	✓			✓		52.1 (− **8.1**)
SAR Guided OPT						
✓		✓			✓	69.2 (− **1.1**)
✓			✓		✓	67.6 (+ **15.7**)
✓				✓	✓	45.5 (+ **4.5**)
	✓	✓			✓	69.3 (− **1.0**)
	✓		✓		✓	67.4 (+ **15.5**)
	✓			✓	✓	47.1 (+ **6.1**)

ignores disparities in learning ability between the teacher and student network. The student network is unaware of the teacher network's learning method and direction. In contrast, in the online distillation method, the student network can learn about the learning process of the teacher network and increase network performance by learning from the teacher's learning abilities.

To illustrate the complementarity and effectiveness of the three modules, we introduce different combinations of CPM, IWM, and OKD for guidance. As shown in Table IV, CPM gets an improvement of 3.4% (from 46.3% to 49.7%) in AP, and OKD gets an improvement of 5.6% (from 46.3% to 51.9%) in AP. IWM is dependent on the output of CPM and therefore cannot be used on its own. This approach (CPM + IWM) fully transfers knowledge from the optical image and gets better performance. When using all three modules for guidance at the same time, our proposed CoLD gets an improvement of 10.4% (from 46.3% to 56.7%) in AP. The results of ablation experiments validate the effectiveness of three different modules from optical images and the capability of our proposed CoLD to fully exploit guidance information. Our CoLD achieves better results by integrating complementary information from optical images.

Our CoLD is the first attempt to adopt logit mimicking to distill localization knowledge for optical images. We combine the probability distribution representation for bounding boxes and the KL-Divergence loss and demonstrate that such a simple logit mimicking technique performs well in improving the distillation performance of object detectors. From Table V, CoLD gets an improvement of 10.4% in AP. KD and DKD are distilled in the classification branch. LD and CoLD are distilled in the regression branch, and the other methods are distilled in both branches. To efficiently transfer information from optical images, it is best to transfer knowledge from both the classification branch and the regression branch (e.g., LD + DKD). To demonstrate that CoLD can transfer classification and localization knowledge using only regression branches, we do four additional sets of experiments (LD + KD, LD + DKD, CoLD + KD, and CoLD + DKD). LD + KD achieves an AP of 55.8% and LD + DKD achieves an AP of 56.2%, which both get significant improvements compared to LD (51.9%). However, compared to CoLD (56.7%), CoLD + KD decreases the AP by 0.7% (from 56.7% to 56.0%) suggesting that CoLD already achieves optimal performance and that introducing KD affects performance instead. CoLD + DKD (56.6%) achieves comparable performance to CoLD, which also indicates that CoLD has already introduced optimal classification knowledge.

3) Effects of Different Parameters on Optical and SAR Images: To evaluate the effect of different parameter images on the performance of the method, we construct a small dataset called optical and SAR images paired dataset with influencing factors such as polarization modes, resolution, and cloud occlusion (OSPRC): it consists of images from Sentinel-1 and Sentinel-2. The SAR images have two different polarization modes and the optical images include 10-m resolution, 20-m resolution, and cloud-occluded images. The dataset is divided into five sets of images, each containing 3300 images of size 256 × 256 pixels, and divided into a training set and a test set in a ratio of 8:2. As shown in Table VI, the results show that

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TABLE VII

COMPARISON WITH STATE-OF-THE-ART ORIENTED OBJECT DETECTION METHODS ON OUR PROPOSED OGSOD-1.0 DATASET WITH SAR IMAGES. C AND S DENOTE CROSS-MODAL METHODS AND SINGLE-MODAL METHODS, RESPECTIVELY. OBB DENOTES ORIENTED BOUNDING BOX

OBB Method	Venue	Backbone	C/S	Parameters	FLOPs	FPS	OGSOD-1.0 (SAR OBB)				
							Oil Tank	Bridge	Harbor	AP_{50}	AP
YOLOv5 OBB (baseline)	-	CSPDarkNet-S	S	**28.55M**	18.15	**65.3**	27.4	66.9	97.4	63.9	31.4
Rotated RepPoints [40]	ICCV'2019	ResNet50	S	36.60M	**12.14**	30.6	53.5	49.1	88.7	63.8	27.8
Rotated FCOS [50]	ICCV'2019	ResNet50	S	31.89M	12.89	35.6	44.0	54.4	90.7	63.0	31.8
S2A-Net [51]	TGRS'2021	ResNet50	S	38.54M	12.28	22.6	46.8	67.0	80.6	64.8	28.3
Oriented R-CNN [52]	ICCV'2021	ResNet50	S	41.13M	26.25	29.9	40.6	68.6	90.2	66.4	34.4
SASM [53]	AAAI'2022	ResNet50	S	36.60M	**12.14**	24.3	**54.2**	50.4	89.9	64.9	30.7
KFIoU [54]	ICLR'2023	ResNet50	S	38.54M	12.28	26.4	30.8	68.7	89.8	63.1	27.0
Rotated RTMDet [55]	ArXiv'2023	CSPNeXt	S	52.26M	12.76	33.7	48.3	54.5	90.4	64.4	31.0
CoLD (Ours)	-	CSPDarkNet-S	C	28.66M	18.80	61.4	31.4	**74.8**	**98.0**	**68.0**	**35.9**

[1] **Best** and *Second best*, Higher AP_{50} and AP are better.

the influence of different polarization modes is small, while the influence of resolution and cloud cover is greater.

In optical image-guided SAR image object detection tasks, different resolutions of optical images can improve the performance of the SAR detector. As shown in Table VI, OPT-10m image and OPT-20m image-guided SAR image object detection increase the AP_{50} by 4.7% and 2.6%, respectively. Interestingly, despite the low-quality of OPT-20m (AP_{50} 41%), it can still significantly improve the performance of SAR-VV object detection. The results of our analysis found that 1532 of the 1891 objects in the SAR-VV predictions and 1653 objects in the OPT-20m predictions intersect. This means that OPT-20m still provides an additional 121 high-quality knowledge for SAR-VV. In contrast, the intersection of SAR-VV and OPT-Cloud (1306) has 1301, indicating that the additional high-quality knowledge provided to SAR-VV by OPT-Cloud is almost negligible. The results show that CoLD can effectively mine high-quality knowledge in optical images.

It is worth noting that as OPT-Cloud contains a lot of low-quality information, it can reduce the performance of the student model. Our proposed IWM module can suppress these low-quality information. As shown in Table VI, the OPT-Cloud-guided SAR-VV decreases the AP_{50} by 7.9% (from 59.4% to 51.5%) without the IWM module. After adding IWM, OPT-Cloud-guided SAR-VV decreases the AP_{50} only 0.9% (from 59.4% to 58.5%). The results show that IWM can effectively suppress low-quality information.

To verify the effectiveness of CoLD in SAR image-guided optical image object detection tasks, we add the experiment of SAR image-guided optical image. As shown in Table VI, SAR-VV-guided OPT-20m increases the AP_{50} by 15.7% (from 51.9% to 67.6%) and SAR-VV-guided OPT-Cloud increases the AP_{50} by 4.5% (from 41.0% to 45.5%). The results show that CoLD is equally effective in SAR-guided optical image object detection tasks.

E. Oriented Object Detection

In the field of remote sensing, rotated labels can better describe some objects with relatively large aspect ratios in images. Therefore, we provide rotated box labels for the dataset OGSOD-1.0 and compare it with some state-of-the-art methods for oriented object detection. As shown in Table VII, our method CoLD performs well under the OGSOD-1.0 (SAR) metric with excellent detection accuracy of 68.0% AP_{50} and 35.9% AP. Compared to YOLOv5 OBB (baseline), our method does not bring a large computational burden, while providing excellent computational efficiency and real-time performance. In contrast, Rotated RepPoints [40] and Rotated fully convolutional one-stage object detection (FCOS) [50] perform well in terms of AP, but they have lower frames/s (FPS) and slower-computational-speed. Single-shot alignment network (S2A-Net) [51] and shape-adaptive selection and measurement (SASM) [53] perform well in terms of both AP and FPS, but have a higher computational burden. Oriented R-CNN [52] performs well in terms of AP_{50} and AP, but has a high-computational burden and requires high hardware devices. Kalman filtering intersection over union (KFIoU) [54] is a novel method to computing IoU that can handle rotated objects better, but performs poorly in AP and FPS. Rotated real-time models for object detection (RTMDet) [55] uses cross stage partial ResNeXt (CSPNeXt), suggesting the next dimension (ResNeXt) as the backbone network, similar to our method, but with a high-computational burden. In summary, our method CoLD has some advantages in terms of performance and computational efficiency, and it can also provide better detection accuracy.

VI. CONCLUSION

In this article, we construct a new benchmark dataset OGSOD-1.0 for optical image-guided SAR image object detection tasks. This is the first optical image-guided SAR image dataset for multiobject detection. On our proposed OGSOD-1.0, we perform extensive experimental analysis of some SAR object detection and cross-modal KD methods. In particular, a novel CoLD method has been designed to progressively transfer object-level knowledge in optical images. The experimental results demonstrate the effectiveness of the proposed CoLD.

In future work, we will investigate strategies to further improve the CoLD architecture by utilizing more advanced techniques, e.g., orientation-sensitive heatmap, making it more applicable to oriented object detection tasks. Moreover, we will further extend the dataset OGSOD-1.0 with the

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addition of more scenarios and environmental factors to inspire further research on this topic.

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